WORKED SOLUTION

STEP II 1987, Question 10

Pure • Group Axioms

> **STEP II 1987, Question 10**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/87-s2-q10

The key idea

Everything happens in the subscripts, so translate each operation into ordinary arithmetic on the indices $1, 2, \ldots, N$: the first operation is "take the maximum" and the second is "take the distance apart, plus one". Then test the four axioms (closure, associativity, identity, inverses) one at a time. A failing axiom needs only one explicit counterexample, and a holding axiom needs a general argument.

Solution

Throughout, write $j \circ k$ for the index of $a_j \circ a_k$, and similarly for $*$. Two elements are equal exactly when their indices are equal.

The operation $\circ$

Here $a_j \circ a_k = a_{\max(j,k)}$.

Closure holds. If $1 \leq j, k \leq N$ then $\max(j,k)$ is one of j and k, so it lies in $\{1, \ldots, N\}$ and $a_j \circ a_k \in S$.

Associativity holds. For any j, k, l,

$$(a_j \circ a_k) \circ a_l = a_{\max(\max(j,k),l)} = a_{\max(j,k,l)} = a_{\max(j,\max(k,l))} = a_j \circ (a_k \circ a_l),$$

since both bracketings give the largest of the three indices.

An identity exists: it is a_1. Every index is at least 1, so $\max(1,k) = \max(k,1) = k$, giving $a_1 \circ a_k = a_k \circ a_1 = a_k$ for every k.

Inverses fail. An inverse of a_j would be some a_k with $a_j \circ a_k = a_1$, that is $\max(j,k) = 1$. But $\max(j,k) \geq j$, so for $j \geq 2$ this is impossible. For example $a_2 \circ a_k$ is a_2 or something later for every k, never a_1. Only a_1 has an inverse (itself).

So S under $\circ$ satisfies closure, associativity and identity, but not inverses, and it is not a group.

Ask yourself Could some other element be the identity, so that the inverse argument has to be redone?

No: an identity is unique. If e and e' were both identities then $e = e \circ e' = e'$. So the only candidate for "$a_j \circ a_k$ equals the identity" is a_1, and the argument above is complete.

The operation $*$

Here $a_j * a_k = a_{|j-k|+1}$. For $N = 4$ the table of indices is

$*$	1	2	3	4
1	1	2	3	4
2	2	1	2	3
3	3	2	1	2
4	4	3	2	1

and the patterns visible in it hold for every N.

Closure holds. If $1 \le j, k \le N$ then $0 \le |j-k| \le N-1$, so $1 \le |j-k|+1 \le N$ and $a_j * a_k \in S$.

An identity exists: it is a_1. Since $k \ge 1$, $|1-k| = k-1$, so $a_1 * a_k = a_k$, and $a_k * a_1 = a_k$ likewise (the operation is commutative because $|j-k| = |k-j|$).

Inverses hold. $a_j * a_j = a_{0+1} = a_1$, the identity, so every element is its own inverse. (This is the diagonal of 1s in the table.)

Associativity fails. Take a_2, a_2, a_3, which needs $N \ge 3$:

$$(a_2 * a_2) * a_3 = a_1 * a_3 = a_3,$$
$$a_2 * (a_2 * a_3) = a_2 * a_{|2-3|+1} = a_2 * a_2 = a_1.$$

These differ, so $*$ is not associative.

So S under $*$ satisfies closure, identity and inverses, but not associativity, and again it is not a group.

Ask yourself Where was $N > 2$ used?

Only in the counterexample to associativity, which uses a_3. When $N = 2$ the table is that of the group of order 2 (a_1 the identity, $a_2 * a_2 = a_1$), and $*$ is associative; for $\circ$ the inverse argument needs only a_2, so $\circ$ fails for $N = 2$ as well.

Checks

Small cases by brute force: for every N from 2 to 9 we listed all triples to test associativity and searched for identities and inverses directly. For $\circ$: closed, associative, identity a_1, and only a_1 invertible. For $*$: closed, identity a_1, every element self-inverse, and

associativity failing (including at (a_2, a_2, a_3)) for every $N \geq 3$ but holding for $N = 2$. You can do the $N = 3$ case by hand from a 3×3 table in a minute.

Common slips

- Claiming $\circ$ has no identity because "max" is not like multiplication. Check the smallest element: a_1 works.
- Trying to prove associativity of $*$ in general, or testing only triples involving a_1 (which always associate, since a_1 is the identity). A single explicit counterexample is what's needed.
- Forgetting closure. It is easy to assume, but here it needs the bound $|j - k| \leq N - 1$.

Beyond STEP

A set with an associative operation and an identity, like S under $\circ$, is called a *monoid*; "max" on a totally ordered set is a standard example, and it is also a *semilattice*. A set with closure, identity and inverses but no associativity, like S under $*$, shows that associativity is a genuinely independent axiom: it cannot be deduced from the other three. For how the group axioms work together, see Allenby, *Rings, Fields and Groups*.

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